

The Complete Effects of Coors Field: Part 1

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Abstract

Coors Field, home to the Colorado Rockies, has long been recognized as Major League Baseball's most hitter-friendly park [1] [2] [3], but its influence on player performance remains incompletely quantified. In this study, we examine how Denver's altitude affects pitching and hitting outcomes by comparing performance at Coors Field with all other parks (AOP). Using Statcast data, we find that the reduced air density at Coors Field's elevation leads to a corresponding decrease in pitch movement, resulting in higher contact rates. When hit in the air, batted balls travel significantly farther which, paired with Coors' unique park dimensions, lead to an increased rate of hits, especially extra-base hits. We later discuss future work on analyzing the additional Coors hangover effect and player evaluations. This study and those that follow will drive strategic decision-making for MLB management with player transactions and optimize player usage and overall team strategy.

Introduction

The goal of this project is to thoroughly analyze the impact Coors Field has on player performance as an exercise. This report is meant to demonstrate how we would communicate the key takeaways of our research. If you would like to see the code used to build this report, as well as additional visuals and additional accounts of our thought process, go to our notebook: https://github.com/skparagas/Coors-Hangover-Effect/blob/main/coors_pt1.ipynb.

For this project we used pitch-level data from Statcast, a tracking tool introduced in Major League Baseball in 2015. The data was primarily sourced from Python's pybaseball library, though in some cases we used data web-scraped directly from the Statcast website. Unless stated otherwise, we used data from the 2015 through 2025 regular season.

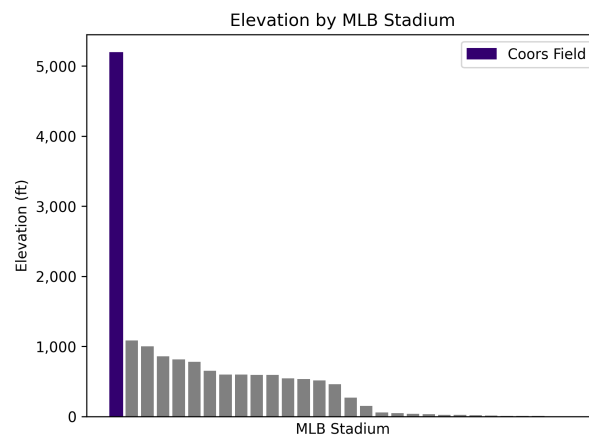


Figure 1. The elevation of all MLB stadiums used in the 2016 season. Other stadiums significantly used from 2015-2025 include Truist Park (1,001 ft), Globe Life Field (545 ft), Salhen Field (521 ft), Sutter Health Park (24 ft), and George M. Steinbrenner Field (34 ft).

Coors Field, the home stadium for the Colorado Rockies baseball team, is located in Denver, Colorado. Nicknamed the "Mile High City", Denver sits roughly 5,200 feet above sea level. As Figure 1 indicates, this elevation is significantly higher than any other Major League city, making it a unique environment for play. Due to the increased altitude, the air is thinner at Coors Field, with an air density roughly 18% lower than it is at sea level [4], though other factors like temperature and humidity have an effect as well.

The Pitch

Pitch Movement

Before getting to the effect Coors Field has on the pitch, it would be worth observing the effect on the baseballs themselves first, but in 2002 the Colorado Rockies installed a humidifier at Coors Field to store baseballs [5]. The humidifier controls moisture to keep the balls from drying out in Denver's high-altitude air. Prior to this, the dry air made baseballs less "slick" and more bouncy, leading to a higher number of home runs and difficulty for pitchers to get a good grip.

Assuming that the humidifier has made the balls in Coors Field indiscernible from the balls in AOP, we first observe how Coors Field affects ball movement out of the pitcher's hand. In conjunction with the standard effects of gravity, pitchers induce additional movement from air resistance, colloquially known as drag, and the Magnus effect, which uses air density to create the force of lift and break based off the direction of the ball's spin [4]. The Magnus force and drag force are both dependent, in fact proportional, to air density, so we would expect to see pitch movement decrease at Coors.

Pitch Type	Count	Horizontal		Vertical	
		Movement	Coors Δ	Movement	Coors Δ
4-Seam Fastball	157	+0.63	-22%	+1.31	-18%
Slider	66	-0.32	-6%	+0.23	-20%
Sinker	60	+1.23	-22%	+0.65	-14%
Changeup	37	+1.05	-21%	+0.60	-19%
Cutter	34	-0.18	-14%	+0.59	-19%
Curveball	24	-0.88	-17%	-0.61	-16%
Knuckle Curve	18	-0.46	-32%	-0.66	-24%
Sweeper	9	-1.22	-13%	+0.23	-67%
Split-Finger	3	+1.04	-26%	+0.25	-3%

Table 1. Induced pitch movement. Movement represents the average amount of inches the given pitch type is induced to move at AOP, to the pitcher's throwing arm side for horizontal and upwards for vertical. Coors Δ represents the average change in that movement at Coors Field. Movement is averaged out for each individual pitcher season with at least 100 pitches of the given pitch type at both Coors and AOP.

Shown in Table 1, pitches do in fact move less at Coors, across all pitch types. There was approximately 18% less induced movement, both horizontally and vertically, which is in line with the decline in air density.

Pitch Location

Unlike pitch movement, which varies across pitchers and pitch types but is relatively consistent at the pitcher and pitch-type level, the location of pitches still varies significantly. Intended location varies depending on factors like count and batter profile. Additionally, execution varies from pitch to pitch, wherein the magnitude and direction of deviation between the intended and actual location can differ drastically. Due to the excessive noise, trends and effects on pitch location can be difficult to observe.

However, pitchers' tend to horizontally locate pitches more towards the direction the pitch is induced to move to [6]. In other words, pitches that move to the right tend to be located more towards the right-side of the plate, and conversely pitches that move to the left tend to be located more towards the left-side of the plate. Given this, we can perform a tentative analysis on this phenomenon and how it's impacted by Coors Field and the reduced pitch movement.

Pitch Type	Right-Handed Pitchers			Left-Handed Pitchers		
	Horiz. Movement	Horiz. Location	Coors Δ	Horiz. Movement	Horiz. Location	Coors Δ
4-Seam Fastball	-0.62	+0.01	—	+0.68	-0.06	—
Sinker	-1.22	-0.24	-19%	+1.24	+0.20	-23%
Slider	+0.43	+0.43	-25%	-0.41	-0.38	-8%
Changeup	-1.13	-0.36	-35%	+1.19	+0.46	-23%
Curveball	+0.82	+0.14	-24%	-0.72	-0.14	-73%
Cutter	+0.23	+0.38	-10%	-0.12	-0.37	-25%
Sweeper	+1.19	+0.48	-9%	-1.19	-0.34	-26%
Split-Finger	-0.90	-0.27	-52%	—	—	—
Knuckle Curve	+0.71	+0.17	-17%	-0.37	-0.15	+1%

Table 2. Horizontal movement and location at AOP and the change in location at Coors. Horizontal location represents the amount of inches the given pitch type is typically located away, horizontally, from the middle of the plate. Neither movement nor location account for handedness; positive amounts represent movement/location in inches to the right from the catcher's perspective, negative amounts represent movement/location in inches to the left from the catcher's perspective. Only pitch types with at least 1,000 pitches at both Coors and AOP are included. Since 4-seam fastballs don't have a standard side of the plate, Coors Δ is left empty.

Table 2 gives further support for the existence of this trend, with the average distance pitches are from the middle of the plate directionally in line with the average movement of those pitches. We see this phenomenon for all pitch types, in both right-handed and left-handed pitchers, except for the 4-seam fastball. The average fastball is horizontally located almost exactly at the middle of plate for both right-handers and left-handers, despite the induced horizontal movement. This may be due to the perceived lack of horizontal movement, since 4-seam fastballs are by far the most common pitch used in baseball and are typically seen as the default pitch. For the other pitch types, we consistently see a decrease at Coors Field in the average distance from the middle of the plate. This appears to be regardless of pitcher handedness or pitch movement, though the magnitude varies greater than what we observed with pitch movement.

Unlike horizontal location, analysis on vertical location did not provide conclusive evidence of any change between Coors Field and AOP. Aside from average location coordinates split by pitch type, there are alternative methods to trying to assess the impact the altitude at Coors Field has on pitch location. One such method, using Statcast's own zone classifications, did not provide conclusive evidence either.

Swing Outcomes

So far our work indicates that at Coors Field pitches consistently have less movement and are horizontally located closer to the middle of the plate. This means batters at Coors tend to face straighter pitches located in more appealing locations.

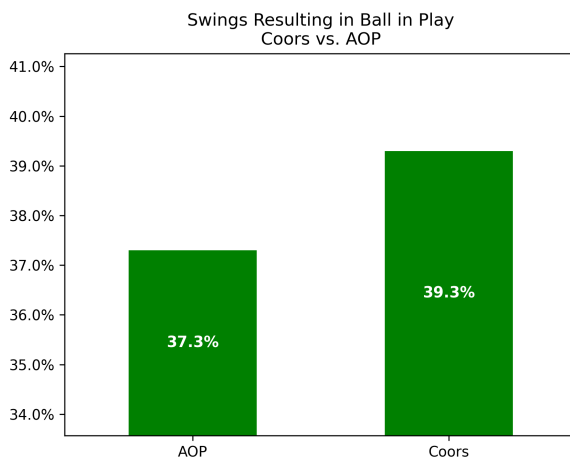


Figure 2. The outcome of swings at Coors vs. AOP.

At Coors Field, swings result in a ball in play roughly 2% more of the time, with a subsequent decrease in the rate of swings that result in missing the ball or fouling it off. This can most likely be attributed to the changes in pitch movement and pitch location, though we don't account for swing tendencies, player ability, or any other possible contributing factors.

The Bat

Exit Velocity and Launch Angle

For a batted ball event (BBE), there are three main factors a batter is in control of: the exit velocity (EV) of the ball leaving the bat, the launch angle (LA) which represents its vertical angle, and spray angle which represents its horizontal angle. Statcast has tracked both EV and LA since 2015, allowing us to analyze the impact Coors Field has on those factors.

With less air resistance at Coors, pitches should retain their velocity more by the time they reach the plate, making it harder for the batter to make contact but easier for them to hit the ball harder upon contact [7], though pitch speed contributes only slightly to EV. Given Coors has roughly 18% less air density and that baseballs lose about 10% of its speed from release to contact [4], pitches would lose 1.8% less of its speed at Coors. Since only 15-25% of exit velocity is attributable to the pitch's speed, we would expect a roughly 0.27-0.45% increase in EV, holding everything else constant. However, analysis of the average exit velocity and overall distribution of EVs at Coors vs. AOP show inconclusive evidence of any change at Coors Field.

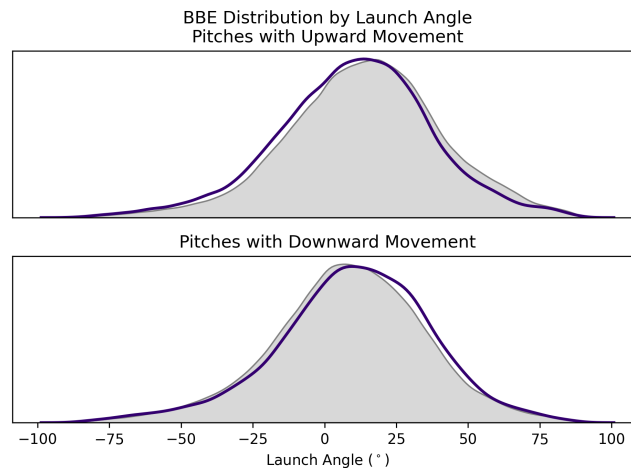


Figure 3. The distribution of launch angles from BBE for pitches with upward-induced movement and pitches with downward-induced movement.

While BBEs at Coors Field have a lower distribution of launch angles than AOP for pitches with upward-induced movement, like fastballs, they have a higher distribution of launch angles for pitches with downward-induced movement, like curveballs. Because of the reduced pitch movement at Coors, pitches like fastballs arrive to the plate lower than they appear to be going and otherwise would have gone. As a result, Figure 3 shows batters tend to get on top of upward movement pitches more at Coors. Conversely, pitches like curveballs end up higher than they appear to be going and would have gone, resulting in batters getting under the ball more. Given that the majority of pitches have upward-induced movement, this phenomenon results in more balls in play with low launch angles (i.e. groundballs).

Hit Distance

After making contact with the bat, baseballs are again in the air and subject to the effects of forces like gravity and air resistance. It is therefore expected that the reduced air resistance at Coors Field has an impact on the flight of a baseball, particularly for those with substantial hang time.

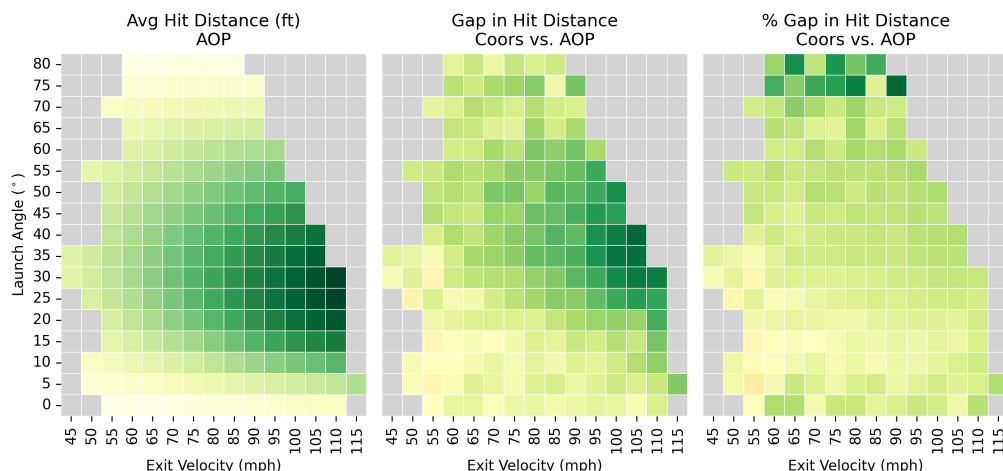


Figure 4. (a) The average hit distance by EV and LA at AOP, (b) the total difference in hit distance at Coors vs. AOP, and (c) the percent change in hit distance at Coors from AOP. Minimum 10 BBEs at both Coors and AOP for each pair of EV and LA buckets for BBEs with a LA of at least 0°.

Controlling for the launch angle and exit velocity of batted balls, BBEs with an upward launch angle consistently travel farther in Colorado. With less drag, baseballs lose less speed before gravity forces them to the ground. The total additive distance at Coors appears to be proportional with the expected distance at AOP as shown in (a) and (b) of Figure 4. However BBEs with higher launch angles and thus more hang time tend to gain a larger proportion of distance.

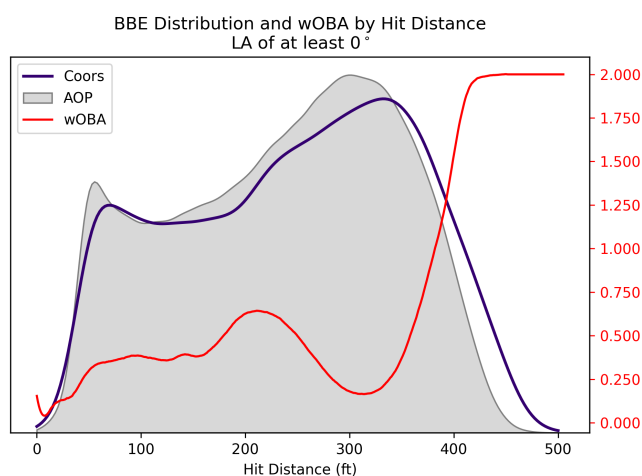


Figure 5. The distribution in the travel distances of BBEs with an upward LA at Coors vs. AOP. Overlaid is the average weighted on-base average (wOBA) for these BBEs in AOP by hit distance.

Within BBEs with an upward launch angle, Figure 5 shows a shift in the distribution of BBE hit distances at Coors vs. AOP, with batted balls trending towards higher hit distances. This shift results in a higher proportion of BBEs in Coors end up going 350+ feet. BBEs at these high distances are considered extremely valuable. As indicated by the average wOBA in AOP, as hit distance increases beyond roughly 315 feet the average offensive value increases dramatically. BBEs with a distance of 350+ feet are shown to have the highest average wOBA of any BBEs hit with a LA of at least 0°.

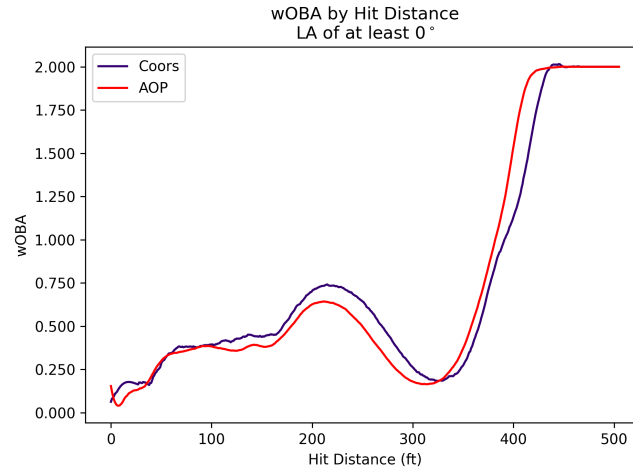


Figure 6. The average wOBA in Coors vs. AOP by hit distance for batted ball events with an upward launch angle.

Although the overall trend in wOBA by hit distance remains the same at Coors Field and the findings from Figure 5 still hold, Figure 6 shows a shift in this trend relative to AOP. Assuming similar distributions in the spray angle of BBEs at Coors vs. AOP, the distance the ball traveled and the time it took to get there appear to be the only significant factors in regards to wOBA. The discrepancy shown in Figure 6 indicates an additional factor to be considered.

The Field

As we have already demonstrated, baseballs carry farther in Coors. Holding everything else constant, this results in more home runs. To help mitigate that effect, Coors Field was built with the largest outfield in Major League Baseball, with outfield fences farther away from home plate than any other MLB stadium.

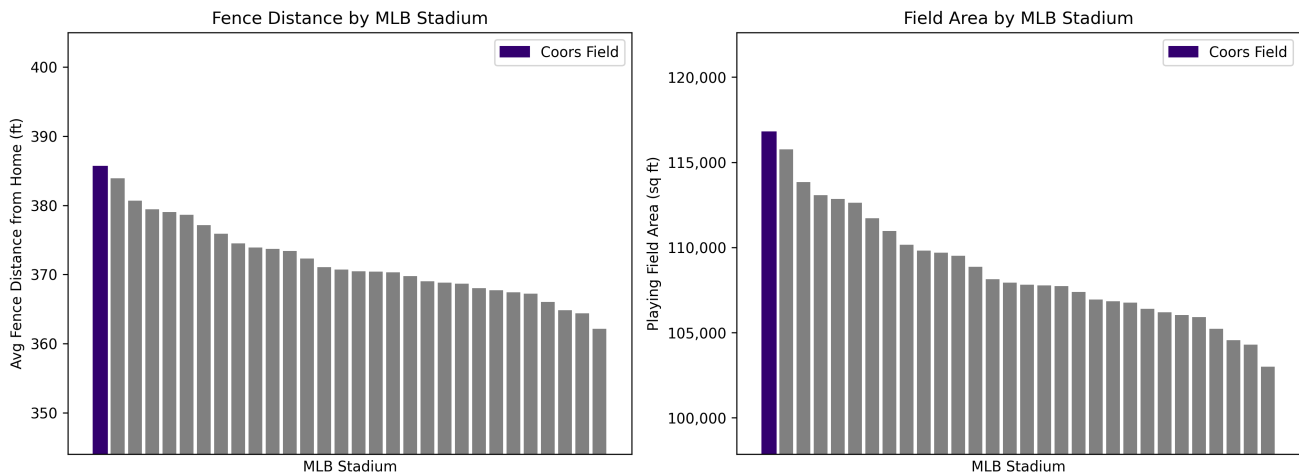


Figure 7. (a) The average fence distance from home plate and (b) the total area of the field in play of all MLB stadiums used in the 2015 season. None of the other stadiums significantly used from 2015-2025, as listed under Figure 1, have had a larger average fence distance or field area than Coors Field.

The outfield fences at Coors are about 15 feet farther out than the average stadium, (a) in Figure 7. It is our hypothesis that outfielders at Coors account for this by positioning themselves differently. Figure 6 shows a local minimum of wOBA

around 315 feet for AOP. We can reasonably assume this is the distance outfielders typically position themselves, making BBEs with that travel distance the easiest for outfielders to reach. At Coors Field, however, the local minimum is around 330 feet. This indicates that outfielders at Coors position themselves farther back in the outfield to compensate for the increased fence distance at Coors Field.

Because of this change in positioning, outfielders start off farther away from BBEs that travel around 200-300 feet at Coors. This allows those BBEs to drop for hits and result in extra bases more often. Conversely, outfielders at Coors start off closer to batted balls that go 330+ feet, so they drop for hits less at Coors.

As the larger outfield was designed to do, BBEs with a distance around 390-420 feet go for home runs less at Coors than the typical park, resulting in an additional gap in average wOBA between Coors and AOP at these distances. However, these batted balls are still valuable at Coors, as having the largest playing field area, per (b) in Figure 7, means outfielders have much a much larger area to cover. Consequentially, batted balls in the outfield more often go for hits, especially extra-base hits like doubles and triples.

Impact Summary

Given the direct and indirect effects Denver's altitude has on pitch movement and location, contact rates, launch angles, hit distance, and even fielder positioning, we quantify the extent these effects have on overall player performance. To prevent introducing biases in the data, we evaluate the performance of away teams at Coors vs. AOP. Since Coors Field is the home park of the Colorado Rockies, performance is heavily weighted by Rockies players, which could affect our results if Rockies hitters or pitchers are significantly different than the rest of MLB players.

	K%	BB%	BABIP	XBH%	HR%	xwOBAcon	wOBAcon	R / PA
AOP	22.7%	7.7%	.277	7.8%	3.1%	.368	.378	.113
Coors	18.7%	8.1%	.310	9.5%	3.4%	.374	.419	.146

Table 3. Performance for away-team batters at Coors vs. AOP.

Based on Table 3, batters at Coors have a reduced strikeout rate. Swings at Coors result in more balls in play, likely due to the reduced pitch movement and more hitter-friendly pitch locations. Interestingly, the decrease in strikeout rate is more than can be attributed to just swing outcomes. The additional drop in K% may be due to changes in batter swing tendencies. If batters increase their swing rates at Coors without decreasing their contact rates, perhaps from awareness of the increased value BBEs have at Coors, that could result in less strikeouts. Another hypothesis is that because of the decreased pitch movement, batters are able to better read pitches and determine if they're balls or strikes. This would allow batters to swing more at pitches in the strike zone but less at pitches out of the strike zone. This could also explain why batters at Coors have an increased walk rate, though the change is minor.

As batting average on balls in play (BABIP) indicates, batted ball events have a much higher batting average at Coors, aided by the increased distance of flyballs and the increased field size. Due to the large outfield, the rate of extra-base hits (XBH) is significantly higher at Coors. With fences farther from home plate than any other MLB stadium, the home run rate at Coors is not notably higher than the average park. The extra carry in the air still gives Coors an above-average HR% but not the highest.

Expected wOBA on contact (xwOBAcon) is fairly similar between Coors and AOP, suggesting that the park-independent quality of contact is comparable. xwOBAcon is slightly higher at Coors, which could be due to the changes in pitch movement and location or because batters tend to have more advantageous batter counts. The difference in strikeout and walk rates indicate that may be the case.

Despite comparable xwOBAcon between Coors and AOP, actual wOBA on contact (wOBAcon) at Coors is significantly higher than it is at AOP. This is supported by the gaps in performance from our other contact metrics like BABIP, XBH%, and HR%. This further confirms that Coors allows better results from BBEs beyond what can be attributed to the quality of contact. With batters having better strikeout and walk rates at Coors, as well as better outcomes from BBEs, the run scoring rate at Coors is approximately 30% higher than AOP.

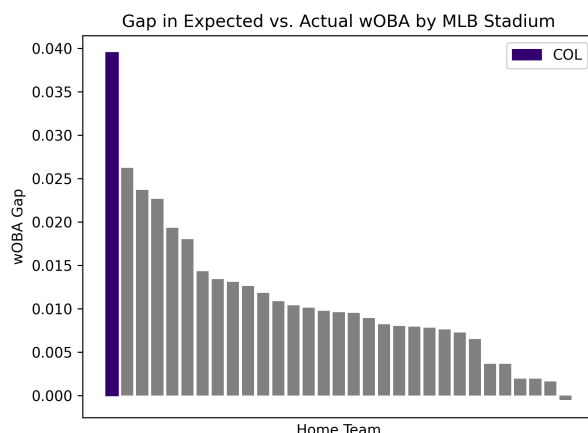


Figure 8. The gap between wOBA and expected wOBA ($wOBA - xwOBA$) by MLB stadium. To account for the different stadiums used by MLB teams from 2015-2025, home teams serve as a proxy. “COL” represents when the home team was the Colorado Rockies, serving as a proxy for when a game was at Coors Field.

Our work has thoroughly demonstrated that Coors Field is unique in several ways, resulting in an environment prone to more offense than the average park. Based on Figure 8, we have substantial evidence that Coors Field is not only hitter-friendly but the most hitter-friendly park in Major League Baseball. MLB players typically outperform their expected wOBA ($xwOBA$), but they do so most at Coors Field, far more than any other park. Additionally, Figure 8 does not account for changes in non-BBE outcomes like walks or strikeouts or in the quality of contact as determined by $xwOBA$. As Table 3 establishes, those factors give Coors Field a further edge in offensive performance.

Due to Coors Field’s extreme elevation and park dimensions, the environment for baseball games there is unlike any other MLB stadium. Pitches move and are placed differently, bats hit pitches differently, and batted balls move differently. These changes allow Coors Field to be far and away the most high-scoring park in Major League Baseball.

Next Steps

While this study covers the general effects Coors Field has on player performance at Coors, there are still many additional topics for future work. Many such topics will be covered in Part 2 of this project.

Our work on BBEs so far have primarily focused on those with a launch angle over 0° , as that is where the impact of air density is most realized, but several results from our research have indicated an effect on BBEs with a launch angle below 0° as well. Further work can be done to determine the existence of this effect, as well as the source of it. It is possible that Coors impacts how groundballs move off the bat as well as the fielders attempting to make a play on them.

Additionally, our research focused on the existence and overall magnitude of the effects of Coors Field. Future work can investigate how these effects change based off various factors, including time and player profile. It is more than likely that some of the effects of Coors on players change over time across a season, especially with temperature being a significant factor in air density, as well as across multiple seasons. Also, given how many different approaches to baseball there are, certain batters and pitchers may be better suited to utilizing or mitigating the effects of Coors and high-altitude environments. This could be of particular importance to teams with a significant number of games in such environments, allowing them to optimize decision-making in regards to player transactions and usage.

The primary focus of Part 2 of this project will cover the phenomenon known as the “Coors hangover effect” [8]. This pertains to the additional effect Coors Field has, mainly on the Colorado Rockies, away from Coors. Similar to the concept of sea legs, players have to readjust their approaches when playing at lower elevations, causing poor performance.

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